

- Q.27 Write a short note on measuring and reference electrode used in pH measurement.
- Q.28 Explain the working of photoelectric tacho generator with diagram.
- Q.29 Explain how a resistor is used to measure the thickness of an object.
- Q.30 How does clinometer help in angle measurement?
- Q.31 Explain strain gauge accelerometer with diagram.
- Q.32 Draw and explain the functional block diagram of measurement system.
- Q.33 Explain micrometer and meter rule for length measurement.
- Q.34 What are the methods used for linear velocity measurement? Explain one of them.
- Q.35 Write a short note on photovoltaic cell.

SECTION-D

Note: Long answer type questions. Attempt any two questions out of three questions. (2x10=20)

- Q.36 Write a short note on the following
- | | |
|---------------------|---------------|
| a) led | b) LDR |
| c) Photo transistor | c) Meter rods |
| d) Tapes | |
- Q.37 What are the area measuring methods? Explain them all with proper diagram.
- Q.38 What is measurement system? Draw and explain the elements of a generalized measurement system. Also explain the input- output configuration of instrumentation system.

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6th Sem / Instrumentation & Control Sub : Advanced Measurement Techniques

Time : 3Hrs.

M.M. : 100

SECTION-A

Note: Multiple choice questions. All questions are compulsory (10x1=10)

- Q.1 A seismic vibration measurement instrument consists of
- | | |
|-----------|---------------------|
| a) Mass | b) Spring |
| c) Damper | d) All of the above |
- Q.2 Ultrasonic method is used for the measurement of _____.
- | | |
|--------------|-----------------------|
| a) Thickness | b) Vibrations |
| c) Viscosity | d) pH of the solution |
- Q.3 Which of the following instruments is used to measure the angular velocity?
- | |
|------------------------------------|
| a) Doppler Transducer |
| b) LVDT |
| c) Tooth rotor variable reluctance |
| d) Potentiometer |
- Q.4 Which of the following is not used for angular velocity measurement?
- | |
|-------------------------------------|
| a) Stroboscope |
| b) Photoelectric tacho generator |
| c) Tooth rotor wheel tachogenerator |
| d) None of the above |

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- Q.5 Which of the following instrument is used to measure angle?
 a) Vernier Caliper b) Clinometers
 c) Numerical method d) Meter rods
- Q.6 LED stands for
 a) Light emitting decoder
 b) Light emitting device
 c) Light emitting diode
 d) Light ejecting diode
- Q.7 A solution is acidic in nature. Which of the following will be the pH of solution?
 a) 2 b) 16
 c) 7 d) 14
- Q.8 A falling piston method is used in the measurement of
 a) Angle b) Area
 c) pH d) Viscosity
- Q.9 A/an _____ is an electronic component that transfers signals using optical path.
 a) Photo conductor b) Opto-isolator
 c) Photo thyristor d) LDR
- Q.10 The input first touches the _____ block in a measurement system.
 a) Variable conversion element
 b) Data representation element
 c) Data transmission element
 d) Primary sensing element

SECTION-B

Note: Objective type questions. All questions are compulsory. (10x1=10)

- Q.11 Graphical method is used for the _____ measurement.

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- Q.12 A protractor can measure angle up to 360^0 . (True / False)
- Q.13 A hydrometer is used to measure _____.
- Q.14 Expand LDR.
- Q.15 Define length.
- Q.16 A micrometer is used for the measurement of length. (True/ False)
- Q.17 _____ is a semiconductor device that converts sunlight directly into electricity.
- Q.18 Define pH of a solution.
- Q.19 Electromagnetic type linear velocity measurement method works on the principle of _____.
- Q.20 in LVDT type seismic vibration transducer, _____ is used as the seismic mass.

SECTION-C

Note: Short answer type questions. Attempt any twelve questions out of fifteen questions. (12x5=60)

- Q.21 Write a short note on falling sphere viscometer.
- Q.22 What are the tools used for length measurement? Explain any one of them.
- Q.23 Draw and explain the input-output configuration of instrument system.
- Q.24 Write a short note on graphical method of area measurement.
- Q.25 What are the light modulating techniques? Explain light attenuation.
- Q.26 Explain force balance method of density measurement.

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